

## JEE-MAINS-2020-JAN-9-SHIFT-2

### PHYSICS

**Question 1:** A particle starts from the origin at  $t = 0$  with an initial velocity of  $3.0 \hat{i}$  m/s and moves in the x-y plane with a constant acceleration  $(6.0 \hat{i} + 4.0 \hat{j})$  m/s<sup>2</sup>. The x-coordinate of the particle at the instant when its y-coordinate is 32m is D meters. The value of D is :

**Options:**

- (a) 50
- (b) 40
- (c) 32
- (d) 60

**Question 2:** A spring mass system (mass  $m$ , spring constant  $k$  and natural length  $l$ ) rests in equilibrium on a horizontal disc. The free end of the spring is fixed at the centre of the disc. If the disc together with spring mass system, rotates about its axis with an angular velocity  $\omega$ , ( $k \gg m\omega^2$ ) the relative change in the length of the spring is best given by the

**Options:**

- (a)  $\sqrt{\frac{2}{3}} \left( \frac{m\omega^2}{k} \right)$
- (b)  $\frac{m\omega^2}{3k}$
- (c)  $\frac{2m\omega^2}{3k}$
- (d)  $\frac{m\omega^2}{k}$

**Question 3:** A rod of length  $L$  has non-uniform linear mass density given by

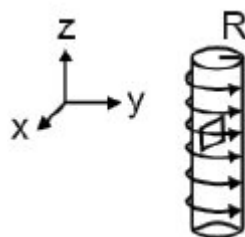
$\rho(x) = a + b \left( \frac{x}{L} \right)^2$  where  $a$  and  $b$  are constants and  $0 \leq x \leq L$ . The value of  $x$  for the centre of mass of the rod is at :

**Options:**

- (a)  $\frac{4}{3} \left( \frac{a+b}{2a+3b} \right) L$
- (b)  $\frac{3}{4} \left( \frac{2a+b}{3a+b} \right) L$
- (c)  $\frac{3}{2} \left( \frac{2a+b}{3a+b} \right) L$
- (d)  $\frac{3}{2} \left( \frac{a+b}{2a+b} \right) L$

**Question 4:** An electron gun is placed inside a long solenoid of radius  $R$  on its axis. The solenoid has  $n$  turns/length and carries a current  $I$ . The electron gun shoots an electron along

the radius of the solenoid with speed  $v$ . If the electron does not hit the surface of the solenoid, maximum possible value of  $v$  is (all symbols have their standard meaning)



**Options:**

- (a)  $\frac{e\mu_0 nIR}{2m}$
- (b)  $\frac{e\mu_0 nIR}{m}$
- (c)  $\frac{2e\mu_0 nIR}{m}$
- (d)  $\frac{e\mu_0 nIR}{4m}$

**Question 5:** A particle of mass  $m$  is projected with a speed  $u$  from the ground at an angle  $\theta = \pi/3$  w.r.t. horizontal ( $x$ -axis). When it has reached its maximum height, it collides completely inelastically with another particle of the same mass and velocity  $u\hat{i}$ . The horizontal distance covered by the combined mass before reaching the ground is:

**Options:**

- (a)  $\frac{3\sqrt{3}}{8} \frac{u^2}{g}$
- (b)  $2\sqrt{2} \frac{u^2}{g}$
- (c)  $\frac{5}{8} \frac{u^2}{g}$
- (d)  $\frac{3\sqrt{2}}{4} \frac{u^2}{g}$

**Question 6:** A wire of length  $L$  and mass per unit length  $6.0 \times 10^{-3} \text{ kgm}^{-1}$  is put under tension of 540 N. Two consecutive frequencies that it resonates at are 420 Hz and 490 Hz. Then  $L$  in meters is :

**Options:**

- (a) 8.1 m
- (b) 2.1 m
- (c) 11.1 m
- (d) 5.1 m

**Question 7:** Two gases-argon (atomic radius 0.07 nm, atomic weight 40) and xenon (atomic radius 0.1 nm, atomic weight 140) have the same number density and are at the same temperature. The ratio of their respective mean free times is closest to :

**Options:**

- (a) 1.83
- (b) 2.3
- (c) 3.67
- (d) 1.07

**Question 8:** For the four sets of three measured physical quantities as given below. Which of the following options is correct?

- (i)  $A_1 = 24.36$ ,  $B_1 = 0.0724$ ,  $C_1 = 256.2$
- (ii)  $A_2 = 25.44$ ,  $B_2 = 16.082$ ,  $C_2 = 240.2$
- (iii)  $A_3 = 25.2$ ,  $B_3 = 19.2812$ ,  $C_3 = 236.183$
- (iv)  $A_4 = 25$ ,  $B_4 = 236.191$ ,  $C_4 = 19.5$

**Options:**

- (a)  $A_1 + B_1 + C_1 < A_3 + B_3 + C_3 < A_2 + B_2 + C_2 < A_4 + B_4 + C_4$
- (b)  $A_1 + B_1 + C_1 = A_2 + B_2 + C_2 = A_3 + B_3 + C_3 = A_4 + B_4 + C_4$
- (c)  $A_4 + B_4 + C_4 < A_1 + B_1 + C_1 = A_2 + B_2 + C_2 = A_3 + B_3 + C_3$
- (d)  $A_4 + B_4 + C_4 < A_2 + B_2 + C_2 < A_1 + B_1 + C_1 = A_3 + B_3 + C_3$

**Question 9:** A uniformly thick wheel with moment of inertial I and radius R is free to rotate about its centre of mass (see figure). A massless string is wrapped over its rim and two blocks of masses  $m_1$  and  $m_2$  ( $m_1 > m_2$ ) are attached to the ends of the string. The system is released from rest. The angular speed of the wheel when  $m_1$  descends by a distance h is :

**Options:**

- (a)  $\left[ \frac{m_1 + m_2}{(m_1 + m_2)R^2 + 1} \right]^{\frac{1}{2}} gh$
- (b)  $\left[ \frac{2(m_1 + m_2)gh}{(m_1 + m_2)R^2 + 1} \right]^{\frac{1}{2}}$
- (c)  $\left[ \frac{2(m_1 + m_2)gh}{(m_1 + m_2)R^2 + 1} \right]^{\frac{1}{2}}$
- (d)  $\left[ \frac{(m_1 + m_2)gh}{(m_1 + m_2)R^2 + 1} \right]^{\frac{1}{2}} gh$

**Question 10:** The energy required to ionise a hydrogen like ion in its ground state is 9 Rydbergs. What is the wavelength of the radiation emitted when the electron in this ion jumps from the second excited state to the ground state

**Options:**

- (a) 24.2 nm
- (b) 6.8 nm

- (c) 35.8 nm  
(d) 11.4 nm

**Question 11:** A small circular loop of conducting wire has radius  $a$  and carries current  $I$ . It is placed in a uniform magnetic field  $B$  perpendicular to its plane such that when rotated slightly about its diameter and released, it starts performing simple harmonic motion of time period  $T$ . If the mass of the loop is  $m$  then:

**Options:**

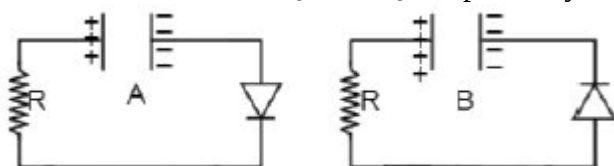
- (a)  $T = \sqrt{\frac{\pi m}{IB}}$   
(b)  $T = \sqrt{\frac{2m}{IB}}$   
(c)  $T = \sqrt{\frac{\pi m}{2IB}}$   
(d)  $T = \sqrt{\frac{2\pi m}{IB}}$

**Question 12:** An electron of mass  $m$  and magnitude of charge  $|e|$  initially at rest gets accelerated by a constant electric field  $E$ . The rate of change of de-Broglie wavelength of this electron at time  $t$  ignoring relativistic effects is

**Options:**

- (a)  $-\frac{h}{|e|Et}$   
(b)  $-\frac{h}{|e|E\sqrt{t}}$   
(c)  $\frac{-h}{|e|Et^2}$   
(d)  $\frac{|e|Et}{h}$

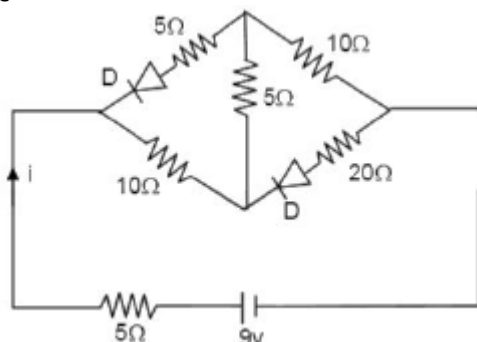
**Question 13:** Two identical capacitors A and B, charged to the same potential 5V are connected in two different circuits as shown below at time  $t = 0$ . If the charge on capacitors A and B at time  $t = CR$  is  $Q_A$  and  $Q_B$  respectively, then (Here  $e$  is the base of natural logarithm)



**Options:**

- (a)  $Q_A = VC$ ,  $Q_B = CV$   
(b)  $Q_A = VC$ ,  $Q_B = VC/e$   
(c)  $Q_A = VC/e$ ,  $Q_B = CV/2$   
(d)  $Q_A = CV/2$ ,  $Q_B = VC/e$

**Question 14:** The current  $i$  in the network is :



**Options:**

- (a) 0.2 A
- (b) 0.3 A
- (c) 0 A
- (d) 0.6 A

**Question 15:** Planet A has mass  $M$  and radius  $R$ . Planet B has half the mass and half the radius of Planet A. If the escape velocities from the Planets A and B are  $V_A$  and  $V_B$ , respectively, then  $V_A/V_B = n/4$ . The value of  $n$  is:

**Options:**

- (a) 3
- (b) 2
- (c) 1
- (d) 4

**Question 16:** Two steel wires having same length are suspended from a ceiling under the same load. If the ratio of their energy stored per unit volume is 1 : 4, the ratio of their diameters is :

**Options:**

- (a) 1 : 2
- (b)  $1:\sqrt{2}$
- (c) 2 : 1
- (d)  $\sqrt{2}:1$

**Question 17:** There is a small source of light at some depth below the surface of water (refractive index =  $4/3$ ) in a tank of large cross sectional surface area. Neglecting any reflection from the bottom and absorption by water, percentage of light that emerges out of surface is (nearly) : [Use the fact that surface area of a spherical cap of height  $h$  and radius of curvature  $r$  is  $2\pi rh$ ]

**Options:**

- (a) 50%
- (b) 34%
- (c) 17%
- (d) 21%

**Question 18:** A plane electromagnetic wave is propagating along the direction  $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$  with its polarization along the direction  $\hat{k}$ . The correct form of the magnetic field of the wave would be (here  $B_0$  is an appropriate constant) :

**Options:**

- (a)  $B_0 \frac{\hat{i} + \hat{j}}{\sqrt{2}} \cos \left( \omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}} \right)$
- (b)  $B_0 \frac{\hat{i} - \hat{j}}{\sqrt{2}} \cos \left( \omega t + k \frac{\hat{i} + \hat{j}}{\sqrt{2}} \right)$
- (c)  $B_0 \frac{\hat{i} - \hat{j}}{\sqrt{2}} \cos \left( \omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}} \right)$
- (d)  $B_0 \hat{k} \cos \left( \omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}} \right)$

**Question 19:** In LC circuit the inductance  $L = 40$  mH and capacitance  $C = 100$   $\mu$ F. If a voltage  $V(t) = 10 \sin(314t)$  is applied to the circuit, the current in the circuit is given as:

**Options:**

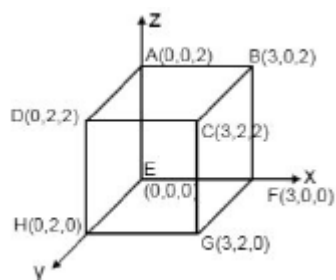
- (a)  $0.52 \cos 314 t$
- (b)  $10 \cos 314 t$
- (c)  $5.2 \cos 314 t$
- (d)  $0.52 \sin 314 t$

**Question 20:** A small spherical droplet of density  $d$  is floating exactly half immerse in a liquid of density  $\rho$  and surface tension  $T$ . The radius of the droplet is (take note that the surface tension applies an upward force on the droplet):

**Options:**

- (a)  $r = \sqrt{\frac{T}{(d + \rho)g}}$
- (b)  $r = \sqrt{\frac{3T}{(2d + \rho)g}}$
- (c)  $r = \sqrt{\frac{T}{(d - \rho)g}}$
- (d)  $r = \sqrt{\frac{2T}{3(d + \rho)g}}$

**Question 21:** An electric field  $\vec{E} = 4x\hat{i} - (y^2 + 1)\hat{j}$  N/C passes through the box shown in figure. The flux of the electric field through surface ABCD and BCGF and marked as  $\Phi_I$  and  $\Phi_{II}$  respectively. The difference between  $(\Phi_I - \Phi_{II})$  is (in  $\text{Nm}^2/\text{C}$ ) \_\_\_\_\_.



**Question 22:** Starting at temperature 300 K, one mole of an ideal diatomic gas ( $\gamma = 1.4$ ) is first compressed adiabatically from volume  $V_1$  to  $V_2 = V_1/16$ . It is then allowed to expand isobarically to volume  $2V_2$ . If all the processes are quasi-static then the final temperature of the gas (in  $^{\circ}\text{K}$ ) is (to the nearest integer)\_\_\_\_\_.

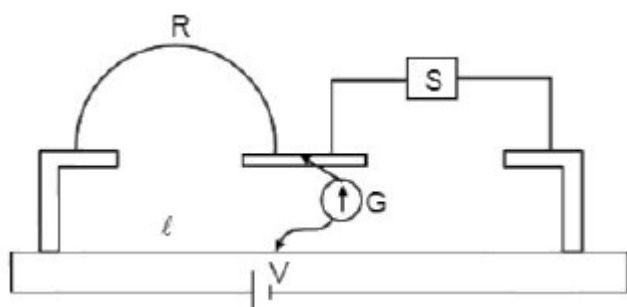
**Question 23:** In a Young's double slit experiment 15 fringes are observed on a small portion of the screen when light of wavelength 500 nm is used. Ten fringes are observed on the same section of the screen when another light source of wavelength  $\lambda$  is used. Then the value of  $\lambda$  is (in nm)

\_\_\_\_\_.

**Question 24:** The circuit shown below is working as a 8V dc regulated voltage source. When 12V is used as input, the power dissipated (in mW) in each diode is ; (considering both zener diodes are identical) \_\_\_\_\_.

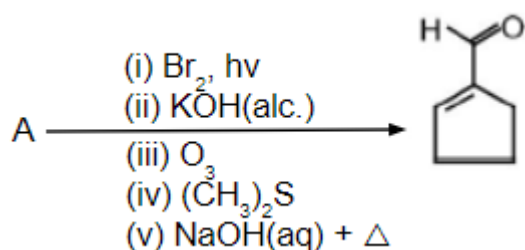
**Question 25:** In a meter bridge experiment  $S$  is a standard resistance.  $R$  is a resistance wire. It is found that balancing length is  $l = 25$  cm. If  $R$  is replaced by a wire of half length and half diameter that of  $R$  of same material, then the balancing distance  $l'$  (in cm) will now be

\_\_\_\_\_.



## CHEMISTRY

**Question 26:** In the following reaction A is:

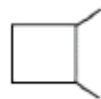


**Options:**

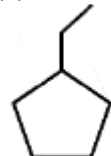
(a)



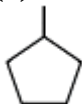
(b)



(c)



(d)



**Question 27:** Amongst the following, the form of water with the lowest ionic conductance at 298 K is:

**Options:**

(a) distilled water

(b) sea water

(c) water from a well

(d) saline water used for intravenous injection

**Question 28:** Among the statements (a)–(d), the correct ones are :

(a) Lithium has the highest hydration enthalpy among the alkali metals.

(b) Lithium chloride is insoluble in pyridine.

(c) Lithium cannot form ethynide upon its reaction with ethyne.

(d) Both lithium and magnesium react slowly with  $H_2O$ .

**Options:**

(a) (a), (b) and (d) only

(b) (b) and (c) only

(c) (a) and (d) only

(d) (a), (c) and (d) only

**Question 29:** Which polymer has 'chiral' monomer(s)?

**Options:**

(a) Buna-N

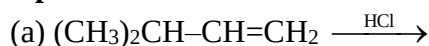
(b) PHBV

(c) Neoprene

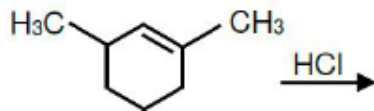
(d) Nylon 6, 6

**Question 30:** Which of the following reactions will not produce a racemic product?

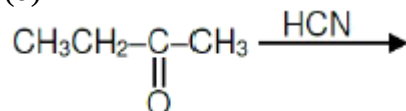
**Options:**



(b)

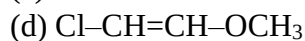
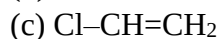
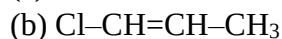
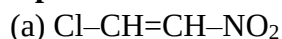


(d)

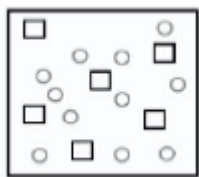


**Question 31:** Which of the following has the shortest C–Cl bond?

**Options:**



**Question 32:** In the Figure shown below reactant A (represented by square) is in equilibrium with product B (represented by circle). The equilibrium constant is:



**Options:**

(a) 4

(b) 8

(c) 1

(d) 2

**Question 33:** The number of  $\text{sp}^2$  hybrid orbitals in a molecule of benzene is:

**Options:**

(a) 24

(b) 18

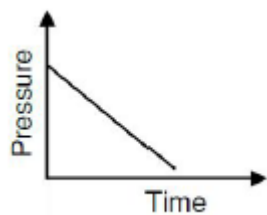
(c) 12

(d) 6

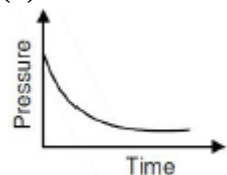
**Question 34:** A mixture of gases  $\text{O}_2$ ,  $\text{H}_2$  and  $\text{CO}$  are taken in a closed vessel containing charcoal. The graph that represents the correct behaviour of pressure with time is:

**Options:**

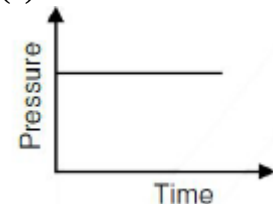
(a)



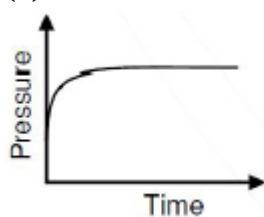
(b)



(c)



(d)



**Question 35:** A, B and C are three biomolecules. The results of the tests performed on them are given below:

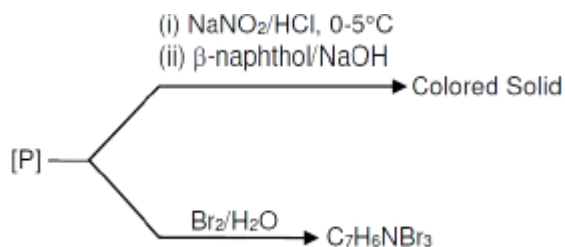
|   | Molisch's Test | Barfoed Test | Biuret Test |
|---|----------------|--------------|-------------|
| A | Positive       | Negative     | Negative    |
| B | Positive       | Positive     | Negative    |
| C | Negative       | Negative     | Positive    |

A, B and C are respectively:

**Options:**

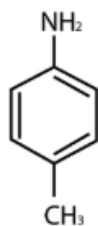
- (a) A = Lactose, B = Glucose, C = Alanine
- (b) A = Glucose, B = Fructose, C = Albumin
- (c) A = Lactose, B = Glucose, C = Albumin
- (d) A = Lactose, B = Fructose, C = Alanine

**Question 36:** Consider the following reactions

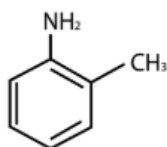


**Options:**

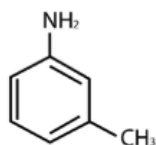
(a)



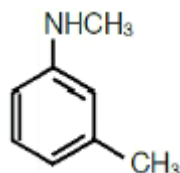
(b)



(c)



(d)

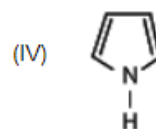
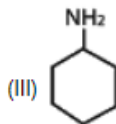
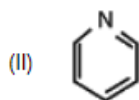
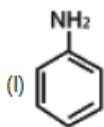


**Question 37:** The first and second ionisation enthalpies of a metal are  $496$  and  $4560 \text{ kJ mol}^{-1}$ , respectively. How many moles of  $\text{HCl}$  and  $\text{H}_2\text{SO}_4$ , respectively, will be needed to react completely with  $1$  mole of the metal hydroxide?

**Options:**

- (a)  $1$  and  $0.5$
- (b)  $2$  and  $0.5$
- (c)  $1$  and  $2$
- (d)  $1$  and  $1$

**Question 38:** The decreasing order of basicity of the following amines is:



**Options:**

- (a) (II) > (III) > (IV) > (I)
- (b) (I) > (III) > (IV) > (II)
- (c) (III) > (II) > (I) > (IV)
- (d) (III) > (I) > (II) > (IV)

**Question 39:** The correct order of the spin-only magnetic moments of the following complexes is :

- (I)  $[\text{Cr}(\text{H}_2\text{O})_6]\text{Br}_2$
- (II)  $\text{Na}_4[\text{Fe}(\text{CN})_6]$
- (III)  $\text{Na}_3[\text{Fe}(\text{C}_2\text{O}_4)_3](\Delta_0 > P)$
- (IV)  $(\text{Et}_4\text{N})_2[\text{CoCl}_4]$

**Options:**

- (a) (III) > (I) > (IV) > (II)
- (b) (III) > (I) > (II) > (IV)
- (c) (I) > (IV) > (III) > (II)
- (d) (II) > (I) > (IV) > (III)

**Question 40:** The reaction of  $\text{H}_3\text{N}_3\text{B}_3\text{Cl}_3$  (A) with  $\text{LiBH}_4$  in tetrahydrofuran gives inorganic benzene (B). Further, the reaction of (A) with (C) leads to  $\text{H}_3\text{N}_3\text{B}_3(\text{Me})_3$ . Compounds (B) and (C) respectively, are:

**Options:**

- (a) Boron nitride and  $\text{MeBr}$
- (b) Borazine and  $\text{MeMgBr}$
- (c) Borazine and  $\text{MeBr}$
- (d) Diborane and  $\text{MeMgBr}$

**Question 41:** The isomer(s) of  $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$  that has/have a  $\text{Cl}-\text{Co}-\text{Cl}$  angle of  $90^\circ$ , is/are:

**Options:**

- (a) cis and trans
- (b) cis only
- (c) meridional and trans
- (d) trans only

**Question 42:** The solubility product of  $\text{Cr}(\text{OH})_3$  at 298 K is  $6.0 \times 10^{-31}$ . The concentration of hydroxide ions in a saturated solution of  $\text{Cr}(\text{OH})_3$  will be:

**Options:**

- (a)  $(2.22 \times 10^{-31})^{1/4}$
- (b)  $(18 \times 10^{-31})^{1/2}$
- (c)  $(4.86 \times 10^{-29})^{1/4}$
- (d)  $(18 \times 10^{-31})^{1/4}$

**Question 43:** The true statement amongst the following is:

**Options:**

- (a) Both  $S$  and  $\Delta S$  are not functions of temperature
- (b)  $S$  is not a function of temperature but  $\Delta S$  is a function of temperature
- (c) Both  $\Delta S$  and  $S$  are functions of temperature
- (d)  $S$  is a function of temperature but  $\Delta S$  is not a function of temperature

**Question 44:** 5g of zinc is treated separately with an excess of

- (a) dilute hydrochloric acid and
- (b) aqueous sodium hydroxide.

The ratio of the volumes of  $H_2$  evolved in these two reactions is:

**Options:**

- (a) 1 : 2
- (b) 2 : 1
- (c) 1 : 4
- (d) 1 : 1

**Question 45:** Biochemical Oxygen Demand (BOD) is the amount of oxygen required (in ppm):

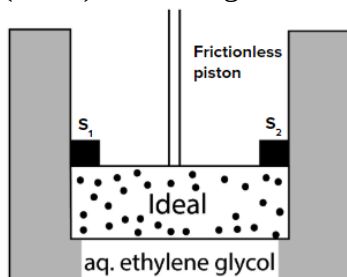
**Options:**

- (a) for the photochemical breakdown of waste present in  $1m^3$  volume of a water body.
- (b) for sustaining life in a water body
- (c) by bacteria to break-down organic waste in a certain volume of a water sample
- (d) by anaerobic bacteria to breakdown inorganic waste present in a water body

**Question 46:** The sum of the total number of bonds between chromium and oxygen atoms in chromate and dichromate ions is \_\_\_\_\_.

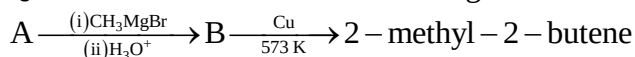
**Question 47:** A sample of milk splits after 60 min. at 300 K and after 40 min. at 400 K when the population of lactobacillus acidophilus in it doubles. The activation energy (in  $kJ\ mol^{-1}$ ) for this process is closest to \_\_\_\_\_. (Given,  $R = 8.3\ J\ mol^{-1}\ K^{-1}$ ,  $\ln(2/3) = 0.4$ ,  $e^{-3} = 4.0$ )

**Question 48:** A cylinder containing an ideal gas (0.1 mol of  $1.0\ dm^3$ ) is in thermal equilibrium with a large volume of 0.5 molal aqueous solution of ethylene glycol at its freezing point. If the stoppers  $S_1$  and  $S_2$  (as shown in the Figure) are suddenly withdrawn, the volume of the gas in litres after equilibrium is achieved will be \_\_\_\_\_. (Given,  $K_f$  (water) =  $2.0\ K\ kg\ mol^{-1}$ ,  $R = 0.08\ dm^3\ atm\ K^{-1}\ mol^{-1}$ )



**Question 49:** 10.30 mg of O<sub>2</sub> dissolved into a liter of sea water of density 1.03 g/mL. The concentration of O<sub>2</sub> in ppm is\_\_\_\_\_.

**Question 50:** Consider the following reactions



The mass percentage of carbon in A is \_\_\_\_\_.

## MATHEMATICS

**Question 51:** Given

$$f(x) = \begin{cases} x & 0 \leq x \leq \frac{1}{2} \\ \frac{1}{2} & x = \frac{1}{2} \\ 1-x & \frac{1}{2} < x \leq 1 \end{cases} \text{ and } g(x) = \left(x - \frac{1}{2}\right)^2, x \in R$$

Then the area (in sq.units) of the region bounded by the curves,  $y = f(x)$  and  $y = g(x)$  between the lines,  $2x = 1$  and  $2x = \sqrt{3}$  is :

**Options:**

- (a)  $\frac{1}{2} - \frac{\sqrt{3}}{4}$   
 (b)  $\frac{1}{3} + \frac{\sqrt{3}}{4}$   
 (c)  $\frac{1}{2} + \frac{\sqrt{3}}{4}$   
 (d)  $4\frac{\sqrt{3}}{4} - \frac{1}{3}$

**Question 52:** A random variable X has the following probability distribution:

|          |       |      |     |      |        |
|----------|-------|------|-----|------|--------|
| $x_i$    | 1     | 2    | 3   | 4    | 5      |
| $P(x_i)$ | $k^2$ | $2k$ | $k$ | $2k$ | $5k^2$ |

Then  $P(x > 2)$  is equal to :

**Options:**

- (a)  $\frac{23}{36}$   
 (b)  $\frac{7}{12}$   
 (c)  $\frac{1}{36}$

(d)  $\frac{1}{6}$

**Question 53:** If  $z$  be a complex number satisfying  $|\operatorname{Re}(z)| + |\operatorname{Im}(z)| = 4$ , then  $|z|$  cannot be:

**Options:**

(a)  $\sqrt{10}$

(b)  $\sqrt{8}$

(c)  $\sqrt{7}$

(d)  $\frac{\sqrt{17}}{2}$

**Question 54:** Let  $a, b \in \mathbb{R}$ ,  $a \neq 0$  be such that the equation,  $ax^2 - 2bx + 5 = 0$  has a repeated root  $\alpha$ , which is also a root of the equation,  $x^2 - 2bx - 10 = 0$ . If  $\beta$  is the other root of this equation, then  $\alpha^2 + \beta^2$  is equal to:

**Options:**

(a) 25

(b) 26

(c) 24

(d) 28

**Question 55:** If  $p \rightarrow (p \wedge \sim q)$  is false, then the truth values of  $p$  and  $q$  are respectively:

**Options:**

(a) F, F

(b) F, T

(c) T, F

(d) T, T

**Question 56:** If  $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}; y(1) = 1$

then a value of  $x$  satisfying  $y(x) = e$  is :

**Options:**

(a)  $\sqrt{3}e$

(b)  $\frac{e}{\sqrt{2}}$

(c)  $\sqrt{2}e$

(d)  $\frac{1}{2}\sqrt{3}e$

**Question 57:** The length of the minor axis (along  $y$ -axis) of an ellipse in the standard form is

$\frac{4}{\sqrt{3}}$  If this ellipse touches the line,  $x + 6y = 8$ ; then its eccentricity is :

**Options:**

(a)  $\frac{1}{3}\sqrt{\frac{11}{3}}$

(b)  $\sqrt{\frac{5}{6}}$

(c)  $\frac{1}{2}\sqrt{\frac{11}{3}}$

(d)  $\frac{1}{2}\sqrt{\frac{5}{3}}$

**Question 58:** If  $x = 2\sin\theta - \sin 2\theta$  and  $y = 2\cos\theta - \cos 2\theta, \theta \in [0, 2\pi]$ , then  $\frac{d^2y}{dx^2}\theta = \pi$  is :

**Options:**

(a)  $-\frac{3}{8}$

(b)  $\frac{3}{2}$

(c)  $\frac{3}{8}$

(d)  $-\frac{3}{4}$

**Question 59:** Let a function  $f : [0, 5] \rightarrow \mathbb{R}$  be continuous,  $f(1) = 3$  and  $F$  be defined as :

$$F(x) = \int_1^x t^2 g(t) \text{ where } g(t) = \int_1^t f(u) du \text{ Then for the function } F, \text{ the point } x = 1 \text{ is}$$

**Options:**

(a) a point of local minima

(b) a point of local maxima

(c) not a critical point

(d) a point of inflection

**Question 60:** Let  $a - 2b + c = 1$ ,

If  $f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$ , then

**Options:**

(a)  $f(-50) = -1$

(b)  $f(50) = 1$

(c)  $f(50) = -501$

(d)  $f(-50) = 501$

**Question 61:** If one end of a focal chord AB of the parabola  $y^2 = 8x$  is at  $A\left(\frac{1}{2}, -2\right)$  then the

equation of the tangent to it at B is :

**Options:**

- (a)  $x - 2y + 8 = 0$
- (b)  $x + 2y + 8 = 0$
- (c)  $2x - y - 24 = 0$
- (d)  $2x + y - 24 = 0$

**Question 62:** Let  $[t]$  denote the greatest integer  $\leq t$  and  $\lim_{x \rightarrow 0} x \left[ \frac{4}{x} \right] = A$

Then the function,  $f(x) = [x^2] \sin(\pi x)$  is discontinuous, when  $x$  is equal to:

**Options:**

- (a)  $\sqrt{A}$
- (b)  $\sqrt{A+1}$
- (c)  $\sqrt{A+5}$
- (d)  $\sqrt{A+21}$

**Question 63:** Let  $a_n$  be the  $n$ th term of a G.P. of positive terms. If

$$\sum_{n=1}^{100} a_{2n+1} = 200 \text{ and } \sum_{n=1}^{100} a_{2n} = 100 \text{ then } \sum_{n=1}^{200} a_n \text{ is equal to}$$

**Options:**

- (a) 175
- (b) 150
- (c) 300
- (d) 225

**Question 64:** If 10 different balls are to be placed in 4 distinct boxes at random, then the probability that two of these boxes contain exactly 2 and 3 balls is :

**Options:**

- (a)  $\frac{965}{2^{11}}$
- (b)  $\frac{965}{2^{10}}$
- (c)  $\frac{945}{2^{10}}$
- (d)  $\frac{945}{2^{11}}$

**Question 65:** Let  $f$  and  $g$  be differentiable functions on  $\mathbb{R}$  such that  $f \circ g$  is the identity function. If for some  $a, b \in \mathbb{R}$ ,  $g'(a) = 5$  and  $g(a) = b$ , then  $f'(b)$  is equal to :

**Options:**

- (a)  $1/5$

- (b)  $\frac{2}{5}$
- (c) 1
- (d) 5

**Question 66:** If  $x = \sum_{n=0}^{\infty} (-1)^n \tan^{2n} \theta$  and  $y = \sum_{n=0}^{\infty} \cos^{2n} \theta$ , for  $0 < \theta < \frac{\pi}{4}$  then

**Options:**

- (a)  $y(1 - x) = 1$
- (b)  $y(1 + x) = 1$
- (c)  $x(1 + y) = 1$
- (d)  $x(1 - y) = 1$

**Question 67:** In the expansion of  $\left( \frac{x}{\cos \theta} + \frac{1}{x \sin \theta} \right)^{16}$  If  $L_1$  is the least value of the term

independent of  $x$  when  $\frac{\pi}{8} \leq \theta \leq \frac{\pi}{4}$  and  $L_2$  is the least value of the term independent of  $x$

when  $\frac{\pi}{16} \leq \theta \leq \frac{\pi}{8}$  then the ratio  $L_2 : L_1$  is equal to:

**Options:**

- (a) 1 : 8
- (b) 1 : 16
- (c) 8 : 1
- (d) 16 : 1

**Question 68:** If  $\int \frac{d\theta}{\cos^2 \theta (\tan 2\theta + \sec 2\theta)} = \lambda \tan \theta + 2 \log_e |f(\theta)| + C$

where  $C$  is a constant of integration, then the ordered pair  $(\lambda, f(\theta))$  is equal to:

**Options:**

- (a)  $(1, 1 + \tan \theta)$
- (b)  $(-1, 1 - \tan \theta)$
- (c)  $(-1, 1 + \tan \theta)$
- (d)  $(1, 1 - \tan \theta)$

**Question 69:** If  $A = \{x \in \mathbb{R} : |x| < 2\}$  and  $B = \{x \in \mathbb{R} : |x - 2| \geq 3\}$ ; then :

**Options:**

- (a)  $A - B = [-1, 2)$
- (b)  $A \cup B = \mathbb{R} - (2, 5)$
- (c)  $B - A = \mathbb{R} - (-2, 5)$
- (d)  $A \cap B = (-2, -1)$

**Question 70:** The following system of linear equations

$$7x + 6y - 2z = 0$$

$$3x + 4y + 2z = 0$$

$$x - 2y - 6z = 0, \text{ has}$$

**Options:**

- (a) infinitely many solution,  $(x, y, z)$  satisfying  $y = 2z$
- (b) infinitely many solution,  $(x, y, z)$  satisfying  $x = 2z$
- (c) Only the trivial solution.
- (d) no solution.

**Question 71:** The number of terms common the two A.P.'s 3, 7, 11, ....., 407 and 2, 9, 16, ....., 709 is \_\_\_\_\_

**Question 72:** If the distance between the plane,  $23x - 10y - 2z + 48 = 0$  and the plane containing the lines  $\frac{x+1}{2} = \frac{y-3}{4} = \frac{z-1}{3}$  and  $\frac{x+3}{2} = \frac{y+2}{6} = \frac{z-1}{\lambda}$  ( $\lambda \in R$ ) is equal to  $\frac{k}{\sqrt{633}}$  then k is equal to \_\_\_\_\_

**Question 73:** Let  $\vec{a}, \vec{b}$  and  $\vec{c}$  be three vectors such that  $|\vec{a}| = \sqrt{3}$ ,  $|\vec{b}| = 5$ ,  $\vec{b} \cdot \vec{c} = 10$  and angle between  $\vec{b}$  and  $\vec{c}$  then  $|\vec{a} \times (\vec{b} \times \vec{c})|$  is equal to \_\_\_\_\_

**Question 74:** If the curves,  $x^2 - 6x + y^2 + 8 = 0$  and  $x^2 - 8y + y^2 + 16 - k = 0$ , ( $k > 0$ ) touch each other at a point, then the largest value of k is \_\_\_\_\_

**Question 75:** If  $C_r = {}^{25}C_r$  and  $C_0 + 5.C_1 + 9.C_2 + \dots + (101).C_{25} = 2^{25}.k$ , then k is equal to \_\_\_\_\_

**Answer Key:**

|                       |                        |                       |                       |                       |                |                      |                      |                       |                       |
|-----------------------|------------------------|-----------------------|-----------------------|-----------------------|----------------|----------------------|----------------------|-----------------------|-----------------------|
| <b>1. (d)</b>         | <b>2. (d)</b>          | <b>3. (b)</b>         | <b>4. (b)</b>         | <b>5. (a)</b>         | <b>6. (d)</b>  | <b>7. (d)</b>        | <b>8. (d)</b>        | <b>9. (c)</b>         | <b>10. (d)</b>        |
| <b>11. (b)</b>        | <b>12. (d)</b>         | <b>13. (d)</b>        | <b>14. (b)</b>        | <b>15. (a)</b>        | <b>16. (c)</b> | <b>17. (b)</b>       | <b>18. (b)</b>       | <b>19. (c)</b>        | <b>20. (a)</b>        |
| <b>21.</b><br>(40.00) | <b>22.</b><br>(750.00) | <b>23.</b><br>(1819)  | <b>24.</b><br>(48.00) | <b>25.</b><br>(40.00) | <b>26. (a)</b> | <b>27. (a)</b>       | <b>28. (d)</b>       | <b>29. (b)</b>        | <b>30. (a)</b>        |
| <b>31. (a)</b>        | <b>32. (d)</b>         | <b>33. (b)</b>        | <b>34. (b)</b>        | <b>35. (c)</b>        | <b>36. (c)</b> | <b>37. (a)</b>       | <b>38. (c)</b>       | <b>39. (c)</b>        | <b>40. (b)</b>        |
| <b>41. (b)</b>        | <b>42. (d)</b>         | <b>43. (c)</b>        | <b>44. (d)</b>        | <b>45. (c)</b>        | <b>46. 12</b>  | <b>47.</b><br>(3.98) | <b>48.</b><br>(2.17) | <b>49.</b><br>(10.00) | <b>50.</b><br>(66.65) |
| <b>51. (d)</b>        | <b>52. (a)</b>         | <b>53. (c)</b>        | <b>54. (a)</b>        | <b>55. (d)</b>        | <b>56. (a)</b> | <b>57. (c)</b>       | <b>58. (c)</b>       | <b>59. (a)</b>        | <b>60. (b)</b>        |
| <b>61. (a)</b>        | <b>62. (b)</b>         | <b>63. (b)</b>        | <b>64.</b><br>(bonus) | <b>65. (a)</b>        | <b>66. (a)</b> | <b>67. (d)</b>       | <b>68. (c)</b>       | <b>69. (c)</b>        | <b>70. (b)</b>        |
| <b>71.</b><br>(14.00) | <b>72.</b><br>(3.00)   | <b>73.</b><br>(30.00) | <b>74.</b><br>(36.00) | <b>75.</b><br>(51.00) |                |                      |                      |                       |                       |

## SOLUTIONS:

### PHYSICS

#### **Solution 1:**

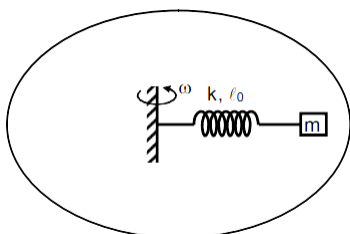
$$\text{From; } S_y = u_y t + \frac{1}{2} a_y t^2$$

$$32 = 0 + \frac{1}{2} \times 4t^2 \Rightarrow t = 4 \text{ sec}$$

$$\text{Now, } S_x = u_x t + \frac{1}{2} a_x t^2$$

$$= 3 \times 4 + \frac{1}{2} \times 6 \times 16 = 16 \text{ m.}$$

#### **Solution 2:**



Here,

$$m\omega^2(\ell_0 + x) = kx$$

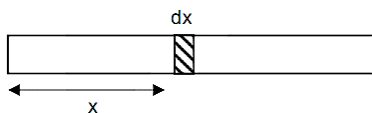
$$\left(\frac{\ell_0}{x} + 1\right) = \frac{k}{m\omega^2}$$

$$x = \frac{\ell_0 m\omega^2}{k - m\omega^2}$$

$$\text{As, } k \gg m\omega^2$$

$$\text{So, } \frac{x}{\ell_0} \text{ is equal to } \frac{m\omega^2}{k}.$$

#### **Solution 3:**



Centre of mass can be located as below:

$$x_{cm} = \frac{1}{M} \int_0^\ell x.dM$$

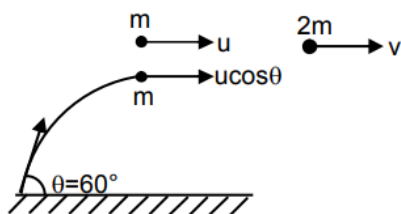
$$dM = \lambda.dx = \left(a + b\left(\frac{x}{\ell}\right)^2\right).dx$$

$$\begin{aligned}
 x_{cm} &= \frac{\int x dM}{\int dM} = \frac{\int x \lambda dx}{\int \lambda dx} = \frac{\int_0^\ell x \left( a + \frac{dx^2}{\ell^2} \right) dx}{\int_0^\ell \left( a + \frac{bx^2}{\ell^2} \right) dx} \\
 &= \frac{a \left( \frac{x^2}{2} \right)_0^\ell + \frac{b}{\ell^2} \left( \frac{x^3}{3} \right)_0^\ell}{a \left( x \right)_0^\ell + \frac{b}{\ell^2} \left( \frac{x^3}{3} \right)_0^\ell} \\
 &= \frac{a \frac{\ell^2}{2} + b \frac{\ell^2}{4}}{a \ell + \frac{b \ell}{3}} = \frac{(2a+b) \ell}{(3a+b) 4} \times 3 \\
 &= \frac{3\ell}{4} \left( \frac{2a+b}{3a+b} \right)
 \end{aligned}$$

**Solution 4:**  $R_{\max} = \frac{R}{2} = \frac{mv_{\max}}{e\mu_0 in}$

Or,  $V_{\max} = \frac{Re \mu_0 in}{2m}$

**Solution 5:**



From conservation of momentum;

$$p_i = p_f$$

$$mu + mu \cos \theta = 2mv$$

$$\Rightarrow v = \frac{u(1 + \cos 60^\circ)}{2} = \frac{3}{4}u$$

So horizontal range after collision = vt

$$= v \sqrt{\frac{2H_{\max}}{g}}$$

$$= \frac{3}{4}u \sqrt{\frac{2u^2 \sin^2(60^\circ)}{2g^2}}$$

$$= \frac{3}{4}u^2 \frac{\sqrt{3}}{g} = \frac{3\sqrt{3}u^2}{8g}$$

**Solution 6:**

Fundamental frequency =  $490 - 420 = 70 \text{ Hz}$

$$70 = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$$

$$\Rightarrow 70 = \frac{1}{2\ell} \sqrt{\frac{540}{6 \times 10^{-3}}}$$

$$\Rightarrow \ell \approx 2.14 \text{ m}$$

**Solution 7:**

We have: mean free time =  $\frac{1}{\sqrt{2}n\pi d^2 v_{rms}}$

$$\frac{t_{Ar}}{t_{Xe}} = \frac{d_{Xe}^2}{d_{Ar}^2} \times \sqrt{\frac{m_1}{m_2}}$$

$$= \left( \frac{0.1}{0.07} \right)^2 \times \sqrt{\frac{40}{140}} = 1.07$$

**Solution 8:**

We have;

$$A_1 + B_1 + C_1 = 24.36 + 0.0724 + 256.2 = 280.6324 = 281$$

$$A_2 + B_2 + C_2 = 24.44 + 16.082 + 240.2 = 280.722 = 280.7$$

$$A_3 + B_3 + C_3 = 25.2 + 19.2812 + 236.183 = 280.6642 = 281$$

$$A_4 + B_4 + C_4 = 25 + 236.191 + 19.5 = 280.691 = 280$$

$$\text{So, } A_4 + B_4 + C_4 < A_2 + B_2 + C_2 < A_1 + B_1 + C_1 = A_3 + B_3 + C_3$$

**Solution 9:**

Conservation of energy:

$$K_i + U_i = K_f + U_f$$

$$0 + 0 = \frac{1}{2} m_2 v^2 + \frac{1}{2} m_1 v^2 + \frac{1}{2} I \omega^2 - m_1 gh + m_2 gh$$

$$(m_1 - m_2) gh = \frac{1}{2} m_2 (\omega R)^2 + \frac{1}{2} m_1 (\omega R)^2 + \frac{1}{2} I \omega^2$$

$$\sqrt{\frac{2(m_1 - m_2) gh}{\left(m_1 + m_2 + \frac{I}{R^2}\right) R^2}} = \omega$$

$$\frac{1}{R} \sqrt{\frac{2(m_1 - m_2) gh}{\left(m_1 + m_2 + \frac{I}{R^2}\right) R^2}}$$

**Solution 10:**

From frequency condition:

$$h\nu = \frac{hc}{\lambda} = (13.6\text{eV})z^2 \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$n_1 = 1$$

$$n_2 = 3$$

$$\frac{hc}{\lambda} = (13.6\text{eV})(3^2) \left[ \frac{1}{1^2} - \frac{1}{3^2} \right]$$

$$\Rightarrow \frac{hc}{\lambda} = (13.6\text{eV})(9) \left( \frac{8}{9} \right)$$

$$\lambda = \frac{1240}{8 \times 13.6} \text{ nm} = 11.39 \text{ nm}$$

**Solution 11:**

Torque:  $\tau = MB \sin \theta = I\alpha$

$$\pi R^2 I B \theta = \frac{mR^2}{2} \alpha$$

$$\omega = \sqrt{\frac{2\pi IB}{m}} = \frac{2\pi}{T}; \quad T = \sqrt{\frac{2\pi m}{IB}}$$

**Solution 12:**

de Broglie wavelength:

$$\lambda_D = \frac{h}{mv}$$

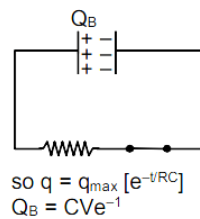
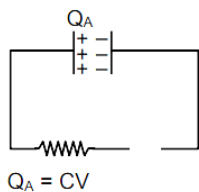
$$v = \frac{eE}{m}t \quad \left( a = \frac{eE}{m} \right)$$

$$\lambda_D = \frac{h}{m \left( \frac{eE}{m} \right) t}$$

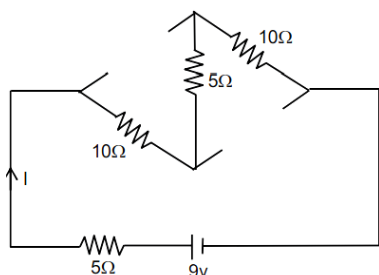
$$\lambda_D = \frac{h}{eEt}$$

$$\frac{d\lambda_d}{dt} = - \frac{h}{|e|Et^2}$$

**Solution 13:** Maximum charge on capacitor = 5 CV



**Solution 14:** Both diodes are in reverse bias  
 Circuit can be redrawn as:



$$I = \frac{9}{30} = \frac{3}{10} A = 0.3A$$

**Solution 15:**

Escape velocity:

$$v_e = \sqrt{\frac{2GM}{R}}$$

$$\therefore \frac{v_1}{v_2} = \frac{\sqrt{2GM}}{R\sqrt{\frac{2GM}{2}} \cdot \frac{1}{R/2}} = 1 = \frac{n}{4}$$

$$\Rightarrow n = 4$$

**Solution 16:**

Potential Energy (U) stored per unit volume;

$$\frac{dU}{dV} = \frac{1}{2} \text{ stress} \times \frac{\text{stress}}{y}$$

$$= \frac{1}{2} \frac{F^2}{A^2 y}$$

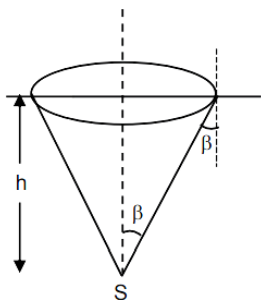
$$\frac{dU}{dV} \propto \frac{1}{d^4}$$

$$\left( \frac{dU}{dV} \right)_1 = \frac{d_2^4}{d_1^4} = \frac{1}{4}$$

$$\frac{d_1}{d_2} = (4)^{1/4}$$

$$\frac{d_1}{d_2} = \sqrt{2} : 1$$

**Solution 17:**



From the figure:

$$\sin \beta = \frac{3}{4}, \cos \beta = \frac{\sqrt{7}}{4}$$

$$\text{Solid angle } d\Omega = 2\pi R^2 (1 - \cos \beta)$$

$$\text{Percentage of light} = \frac{2\pi R^2 (1 - \cos \beta)}{4\pi R^2} \times 100$$

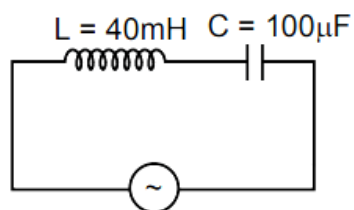
$$= \frac{1 - \cos \beta}{2} \times 100 = \left( \frac{4 - \sqrt{7}}{8} \right) \times 100 \approx 17\%$$

**Solution 18:** EM wave is in direction  $\rightarrow \frac{\hat{i} + \hat{j}}{\sqrt{2}}$

Electric field is in the direction  $\rightarrow \hat{k}$

Direction of propagation of EM wave is in the direction of  $\vec{E} \times \vec{B}$

**Solution 19:**



We have:

$$V = 10 \sin(314t)$$

$$Z = X_C - X_L$$

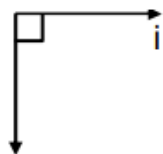
$$= \frac{1}{\omega C} - \omega L$$

$$= \frac{1}{314 \times 100 \times 10^{-6}} - 314 \times 40 \times 10^{-3}$$

$$= 31.84 - 12.56$$

$$= 19.28 \Omega$$

$$X_C > X_L$$



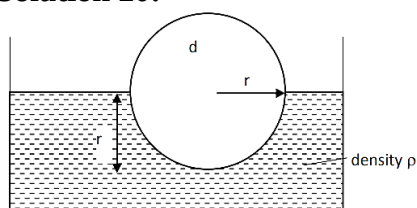
$$V_C - V_L$$

$$i = \frac{V_0}{Z} \sin\left(314t + \frac{\pi}{2}\right)$$

$$\therefore i = \frac{V_0}{Z} \cos(314t) \Rightarrow i = \frac{10}{19.28} \cos(314t)$$

$$\Rightarrow i = 0.52 \cos(314t)$$

### Solution 20:



Balancing the forces:

$$dVg = \rho \left(\frac{V}{2}\right)g + T(2\pi r)$$

$$\Rightarrow d \cdot \frac{4}{3}\pi r^3 g = \rho \cdot \frac{2}{3}\pi r^3 g + 2\pi r T$$

$$\Rightarrow \frac{2}{3}r^2 g (2d - \rho) = 2T$$

$$\Rightarrow r = \sqrt{\frac{3T}{(2d - \rho)g}}$$

### Solution 21: Flux via ABCD

$$\phi_1 = \int \vec{E} \cdot d\vec{A} = 0$$

Flux via BCEF

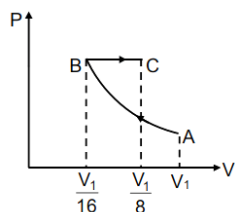
$$\phi_2 = \int \vec{E} \cdot d\vec{A}$$

$$\phi = \vec{E} \cdot \vec{A} = (4x\hat{i} - (y^2 + 1)\hat{j}) \cdot 4\hat{i} = 16x$$

$$\Rightarrow x = 3$$

$$\phi_2 = 48 \text{ Nm}^2 \text{ C}^{-1} ; \quad \phi_1 - \phi_2 = -48 \text{ Nm}^2 \text{ C}^{-1}$$

### Solution 22:



For adiabatic process

$$PV^\gamma = \text{constant}$$

$$TV^{\gamma-1} = \text{constant}$$

$$300(V_1)^{1.4-1} = T_B \left( \frac{V_1}{16} \right)^{2/5}$$

$$T_B = 300 \times 2^{8/5}$$

Now for BC process

$$\frac{V_B}{T_B} = \frac{V_C}{T_C}$$

$$T_C = \frac{V_C T_B}{V_B} = 2 \times 300 \times 2^{8/5}$$

$$T_C = 1818.859$$

$$T_C = 1819 \text{ K}$$

### Solution 23:

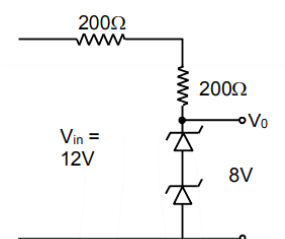
Here we have:

$$15 \times 500 \times \frac{D}{d} = 10 \times \lambda_2 \times \frac{D}{d}$$

$$\lambda_2 = 15 \times 50 \text{ nm}$$

$$\lambda_2 = 750 \text{ nm}$$

### Solution 24:



From the figure:

$$i = \frac{(12-8)}{(200+200)} \text{ A} = \frac{4}{400} = 10^{-2} \text{ A}$$

$$\text{Power loss in each diode} = (4)(10^{-2}) \text{ W} = 40 \text{ mW}$$

**Solution 25:**

We have:

$$\frac{X}{R} = \frac{75}{25} = 3$$

$$R = \frac{\rho \ell}{A} = \frac{4\rho \ell}{\pi d^2}$$

$$R' = \frac{4\rho \left(\frac{\ell}{2}\right)}{\pi \left(\frac{d}{2}\right)^2}$$

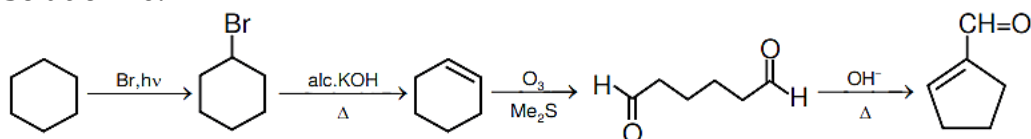
then  $\frac{X}{R'} = \left(\frac{100 - \ell}{\ell}\right)$

$$\frac{100 - \ell}{\ell} = \frac{x}{2R} = \frac{3}{2}$$

$$\ell = 40.00 \text{ cm}$$

**CHEMISTRY**

**Solution 26:**



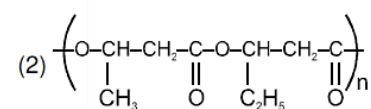
**Solution 27:** Conceptual

**Solution 28:** Conceptual

**Solution 29:**

**Polymers**

(1) Buna-S  $-\left[\text{CH}_2-\underset{\text{Ph}}{\text{CH}}-\text{CH}_2-\text{CH}=\text{CH}-\text{CH}_2\right]-$



PHBV

In PHBV, both monomers have chiral centre.

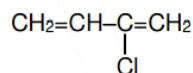
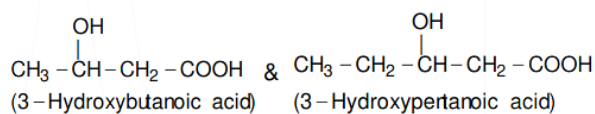
(3) Neoprene  $-\left[\text{CH}_2-\text{CH}=\text{CCl}-\text{CH}_2\right]_n-$

(4) Nylon-6,6

$\sim[\text{CO}(\text{CH}_2)_4\text{CO}-\text{NH}(\text{CH}_2)_6\text{NH}]_n\sim$

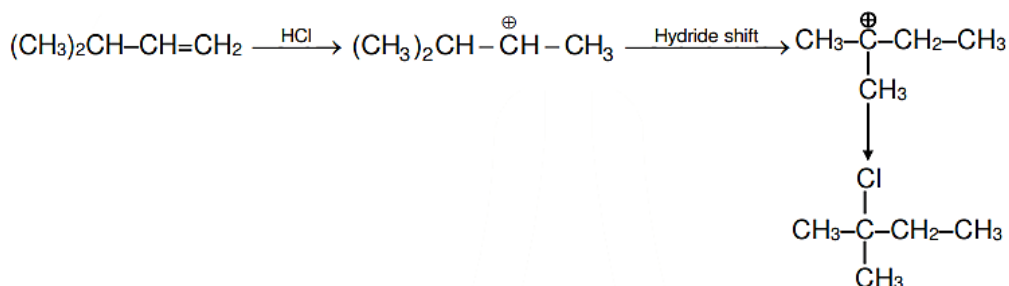
**Monomers**

$\text{CH}_2=\text{CH}-\text{Ph}$  &  $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$



$\text{HO}_2\text{C}-(\text{CH}_2)_4-\text{CO}_2\text{H}$  &  $\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$

**Solution 30:**



Two methyl group are present on the carbon so it's not chiral.

**Solution 31:** Double bond character in carbon-chlorine bond is maximum due to resonance and bond length is shortest.

**Solution 32:** (d)

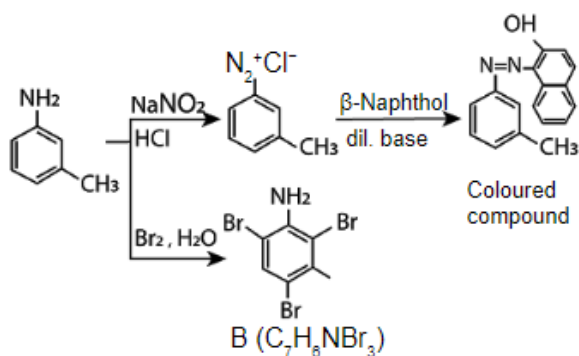
**Solution 33:** In benzene, all carbons are  $\text{sp}^2$  hybrid. Therefore, total  $\text{sp}^2$  hybrid orbitals are 18 in benzene.

**Solution 34:** Conceptual

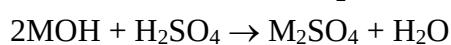
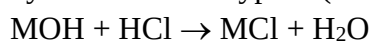
**Solution 35:**

| Lab test        | Molisch's Test             | Barfoed Test | Biuret Test |
|-----------------|----------------------------|--------------|-------------|
| Given by        | Lactose, Glucose, Fructose | Glucose      | Albumin     |
| Do not given by |                            | Fructose     | Alanine     |

**Solution 36:**

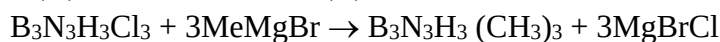
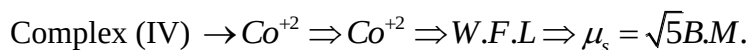
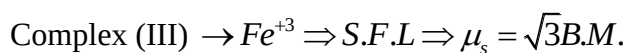
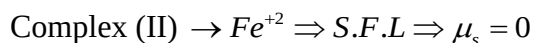
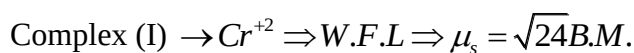


**Solution 37:** This element must belong to group 1 because it is monovalent & form hydroxide of the type  $\text{M}(\text{OH})$ .

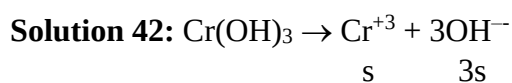
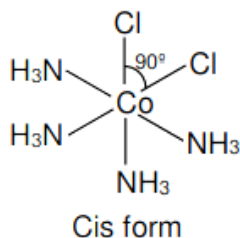


**Solution 38:** Greater the resonance of lone pairs lesser the basic strength.

**Solution 39:**



**Solution 41:**



$$K_{sp} = s.(3s)^3$$

$$\Rightarrow 6 \times 10^{-31} = 27.s^4$$

$$\Rightarrow s = \left( \frac{6}{27} \times 10^{-31} \right)^{1/4}$$

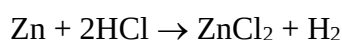
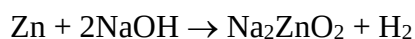
$$[OH^-] = 3s$$

$$= 3 \times \left( \frac{6}{27} \times 10^{-31} \right)^{1/4} = (18 \times 10^{-31})^{1/4} M$$

**Solution 43:**  $\Delta S = \int \frac{dq}{T}$

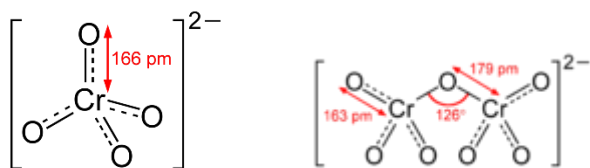
$$S_T = \int_0^T \frac{ncdT}{T}$$

**Solution 44:** Zn metal can react with acid as well as base.



**Solution 45:** Biochemical oxygen demand (BOD) is the amount of dissolved oxygen needed by aerobic biological organisms to break down organic material present in a given water sample at certain temperature over a specific time period.

**Solution 46:**



⇒ Total number of Cr and O bonds in both the ions are 12.

**Solution 47:**  $\ln \frac{K_2}{K_1} = \frac{E_a}{R} \left[ \frac{1}{T_1} - \frac{1}{T_2} \right]$

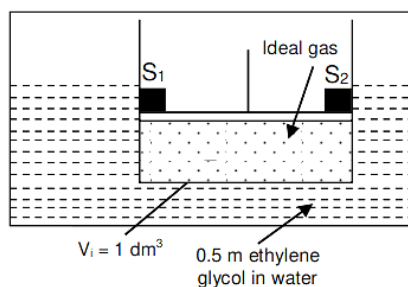
$$\ln \left( \frac{60}{40} \right) = \frac{E_a}{8.3} \times \frac{100}{400 \times 300}$$

$$E_a = \ln \left( \frac{3}{2} \right) \times 8.3 \times 1200$$

$$\Rightarrow E_a = 0.4 \times 8.3 \times 1200$$

$$\Rightarrow E_a = 3.984 \text{ kJ/mol.}$$

**Solution 48:**



$$\Delta T_f = K_f \times m$$

$$= 0.5 \times 2$$

$$T_{\text{initial}} = 272 \text{ K}$$

$$P_{\text{gas}} = \frac{nRT}{V} = \frac{0.1 \times 0.08 \times 272}{1} = 2.176 \text{ atm}$$

After releasing piston, According to Boyle's law

$$P_1 V_1 = P_2 V_2$$

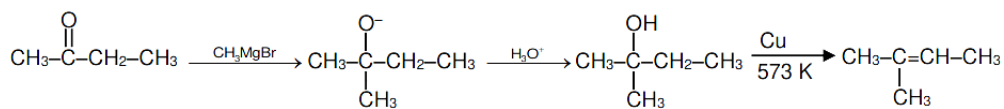
$$2.176 \times 1 = 1 \times V_2$$

$$V_2 = 2.176 \text{ dm}^3 \approx 2.18 \text{ dm}^3$$

**Solution 49:**  $\text{ppm} = (\text{mass of solute} \div \text{mass of solution}) \times 10^6$ .

$$\text{ppm} = \frac{10.3 \times 10^{-3}}{1030} \times 10^6 = 10$$

**Solution 50:**



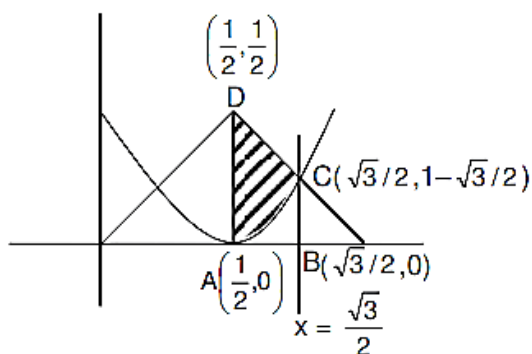
Compound A is  $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\text{CH}_3$  ( $\text{C}_4\text{H}_8\text{O}$ )

$$\% \text{ of Carbon} = 48 \times \frac{100}{72} = 66.67$$

## MATHEMATICS

### Solution 51:

$$\text{Given, } f(x) = \begin{cases} x & 0 \leq x \leq \frac{1}{2} \\ \frac{1}{2} & x = \frac{1}{2} \\ 1-x & \frac{1}{2} < x \leq 1 \end{cases} \text{ and } g(x) = \left(x - \frac{1}{2}\right)^2, x \in R$$



$$\text{Required area} = \text{Area of trapezium } ABCD - \int_{1/2}^{\sqrt{3}/2} \left(x - \frac{1}{2}\right)^2 dx$$

$$= \frac{1}{2} \left( \frac{\sqrt{3}-1}{2} \right) \left( \frac{1}{2} + 1 - \frac{\sqrt{3}}{2} \right) - \frac{1}{3} \left( \left(x - \frac{1}{2}\right)^3 \right)_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}}$$

$$= \frac{\sqrt{3}}{4} - \frac{1}{3}$$

### Solution 52:

Given,

|          |       |      |     |      |        |
|----------|-------|------|-----|------|--------|
| $x_i$    | 1     | 2    | 3   | 4    | 5      |
| $P(x_i)$ | $k^2$ | $2k$ | $k$ | $2k$ | $5k^2$ |

$$\because \sum P(x_i) = 1$$

$$\Rightarrow 6k^2 + 5k = 1$$

$$\Rightarrow 6k^2 + 5k - 1 = 0$$

$$\Rightarrow 6k^2 + 6k - k - 1 = 0$$

$$\Rightarrow (6k - 1)(k + 1) = 0$$

$$\Rightarrow k = -1 (\text{rejected}); k = \frac{1}{6}$$

$$\therefore P(x > 2) = k + 2k + 5k^2$$

$$= \frac{1}{6} + \frac{2}{6} + \frac{5}{36} = \frac{6+12+5}{36} = \frac{23}{36}$$

### Solution 53:

Let  $z = x + iy$

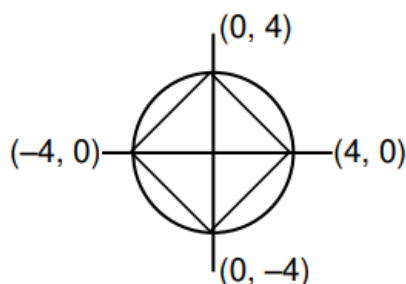
$$\therefore |x| + |y| = 4$$

$$\Rightarrow \text{Minimum value of } |z| = 2\sqrt{2}$$

$$\Rightarrow \text{Maximum value of } |z| = 4$$

$$\Rightarrow |z| \in [\sqrt{8}, \sqrt{16}]$$

So  $|z|$  can't be  $\sqrt{7}$



### Solution 54:

Given,  $\alpha$  is a repeated root of  $ax^2 - 2bx + 5 = 0$

$$\therefore 2\alpha = \frac{2b}{a} \text{ and } \alpha^2 = \frac{5}{a}$$

$$\Rightarrow \alpha = \frac{b}{a} \text{ and } \Rightarrow \frac{b^2}{a^2} = \frac{5}{a}$$

$$\Rightarrow b^2 = 5a \quad \dots\dots(i) \quad (a \neq 0)$$

Given  $\alpha$  and  $\beta$  are the root of the equation  $x^2 - 2bx - 10 = 0$

$$\therefore \alpha + \beta = 2b \quad \dots\dots(ii)$$

$$\text{and } \alpha\beta = -10 \quad \dots\dots(iii)$$

$$\alpha = \frac{b}{a} \text{ is also root of } x^2 - 2bx - 10 = 0$$

$$\Rightarrow b^2 - 2ab^2 - 10a^2 = 0$$

By (i)

$$\Rightarrow 5a - 10a^2 - 10a^2 = 0$$

$$\Rightarrow 20a^2 = 5a$$

$$\Rightarrow a = \frac{1}{4} \text{ and } b^2 = \frac{5}{4}$$

$$\alpha^2 = 20 \text{ and } \beta^2 = 5$$

$$\text{Now } \alpha^2 + \beta^2$$

$$= 5 + 20$$

$$= 25$$

#### Solution 55:

| p | q | $\sim q$ | $p \wedge \sim q$ | $p \rightarrow (p \wedge \sim q)$ |
|---|---|----------|-------------------|-----------------------------------|
| T | T | F        | F                 | F                                 |
| T | F | T        | T                 | T                                 |
| F | T | F        | F                 | T                                 |
| F | F | T        | F                 | T                                 |

#### Solution 56:

$$\text{Given, } \frac{dy}{dx} = \frac{xy}{x^2 + y^2}$$

$$\text{Put } y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{vx^2}{x^2 + v^2 x^2}$$

$$\Rightarrow \frac{1+v^2}{v^3} dv = -\frac{1}{x} dx$$

$$\Rightarrow \int \left( \frac{1}{v^3} + \frac{1}{v} \right) dv = \int \frac{-1}{x} dx$$

$$\Rightarrow \frac{-1}{2} \frac{1}{v^2} + \ln v = -\ln x + c$$

$$\Rightarrow -\frac{x^2}{2y^2} = -\ln y + c$$

$$\text{When } x=1, y=1 \text{ then}$$

$$-\frac{1}{2} = c$$

$$\Rightarrow x^2 = y^2 (1 + 2 \ln y)$$

$$\Rightarrow x^2 = e^2 (3)$$

$$\Rightarrow x = \pm \sqrt{3}e$$

$$\text{So } x = \sqrt{3}e$$

#### Solution 57:

Let standard equation of ellipse is  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\Rightarrow 2b = \frac{4}{\sqrt{3}}$$

$$\Rightarrow b = \frac{2}{\sqrt{3}}$$

Equation of tangent is  $y = mx \pm \sqrt{a^2 m^2 + b^2}$

Comparing with  $y = \frac{-x}{6} + \frac{4}{3}$

$$m = \frac{-1}{6} \text{ and } a^2 m^2 + b^2 = \frac{16}{9}$$

$$\Rightarrow \frac{a^2}{36} + \frac{4}{3} = \frac{16}{9}$$

$$\Rightarrow \frac{a^2}{36} = \frac{16}{9} - \frac{4}{3} = \frac{4}{9}$$

$$\Rightarrow a^2 = 16$$

$$e = \sqrt{1 - \frac{b^2}{a^2}}$$

$$e = \sqrt{1 - \frac{4}{3 \times 16}} = \sqrt{\frac{11}{12}}$$

### Solution 58:

Given,  $x = 2 \sin \theta - \sin 2\theta$  and  $y = 2 \cos \theta - \cos 2\theta$

$$\Rightarrow \frac{dx}{d\theta} = 2 \cos \theta - 2 \cos 2\theta$$

$$\Rightarrow \frac{dy}{d\theta} = -2 \sin \theta + 2 \sin 2\theta$$

$$\therefore \frac{dy}{dx} = \frac{\sin 2\theta - \sin \theta}{\cos \theta - \cos 2\theta}$$

$$= \frac{2 \sin \frac{\theta}{2} \cdot \cos \frac{3\theta}{2}}{2 \sin \frac{\theta}{2} \sin \frac{3\theta}{2}} = \cot \frac{3\theta}{2}$$

$$\frac{d^2 y}{dx^2} = \frac{-3}{2} \operatorname{cosec}^2 \frac{3\theta}{2} \cdot \frac{d\theta}{dx}$$

$$\frac{d^2 y}{dx^2} = \frac{-\frac{3}{2} \operatorname{cosec}^2 \frac{3\theta}{2}}{2(\cos \theta - \cos 2\theta)}$$

$$\left. \frac{d^2 y}{dx^2} \right|_{\theta=\pi} = -\frac{3}{4(-1-1)} = \frac{3}{8}$$

**Solution 59:**

Given,  $F(x) = \int_1^x t^2 g(t)$

$\Rightarrow F'(x) = x^2 g(x)$

$\Rightarrow F'(1) = 1 \cdot g(1) = 0 \quad \dots\dots(1) \quad (\because g(1) = 0)$

Now  $F''(x) = 2xg(x) + x^2 g'(x)$

$\Rightarrow F''(x) = 2xg(x) + x^2 f'(x) \quad (\because g'(x) = f(x))$

$\Rightarrow F''(1) = 0 + 1 \times 3$

$\Rightarrow F''(1) = 3 \quad \dots\dots(2)$

From (1) and (2) F(x) has local minimum at  $x = 1$

**Solution 60:**

Given,  $f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$

Apply  $R_1 \rightarrow R_1 + R_3 - 2R_2$

$\Rightarrow f(x) = \begin{vmatrix} 1 & 0 & 0 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$

$\Rightarrow f(x) = 1$

$\Rightarrow f(50) = 1$

**Solution 61:**

Given parabola is  $y^2 = 8x$

Let  $A\left(\frac{1}{2}, -2\right)$  is  $(2t^2, 4t) \Rightarrow t = \frac{-1}{2}$

Parameter of other end of focal chord is 2

$\Rightarrow$  point is B(8, 8)

$\Rightarrow$  equation of tangent at B is  $8y - 4(x + 8) = 0$

$\Rightarrow 2y - x = 8$

**Solution 62:**

Given,  $\lim_{x \rightarrow 0} x \left[ \frac{4}{x} \right] = A$

$$\Rightarrow \lim_{x \rightarrow 0} x \left( \frac{4}{x} - \left\{ \frac{4}{x} \right\} \right) = A$$

$$\Rightarrow \lim_{x \rightarrow 0} 4 - x \left\{ \frac{4}{x} \right\} = A$$

$$\Rightarrow A = 4$$

Checking from the options, when

$$(A) \ x = \sqrt{A+1} \Rightarrow x = \sqrt{5} \Rightarrow \text{discontinuous}$$

$$(B) \ x = \sqrt{A+21} \Rightarrow x = 5 \Rightarrow \text{continuous}$$

$$(C) \ x = \sqrt{A} \Rightarrow x = 2 \Rightarrow \text{continuous}$$

$$(D) \ x = \sqrt{A+5} \Rightarrow x = 3 \Rightarrow \text{continuous}$$

### Solution 63:

Let GP is  $a, ar, ar^2, \dots$

$$\sum_{n=1}^{100} a_{2n+1} = a_3 + a_5 + \dots + a_{201} = 200 \Rightarrow \frac{ar^2(r^{200}-1)}{r^2-1} = 200 \quad \dots(1)$$

$$\sum_{n=1}^{100} a_{2n} = a_2 + a_4 + \dots + a_{200} = 100 = \frac{ar(r^{200}-1)}{r^2-1} = 100 \quad \dots(2)$$

From (1) and (2), we get  $r = 2$

Adding (1) and (2), we get

$$\Rightarrow a_2 + a_3 + \dots + a_{200} + a_{201} = 300$$

$$\Rightarrow r(a_1 + \dots + a_{200}) = 300$$

$$\sum_{n=1}^{200} a_n = \frac{300}{r} = 150$$

### Solution 64:

**Case-I :** Exactly two box contain 2, 3 balls and other two box does not contains 2, 3 balls equals to

$${}^4C_2 \times 2 \times {}^{10}C_2 \times {}^8C_3 \times \left( 2^5 - \frac{5!}{2!3!} \times 2! \right)$$

$$= 2^4 \times 3^2 \times 45 \times 8 \times 7 = 2^7 \times 3^4 \times 5 \times 7$$

**Case-II :** Two box contains 2 balls each and two box contain 3 balls each equals to

$$\frac{10!}{2!3!2!3!2!2!} \times 4!$$

$$= 2^5 \times 3^3 \times 5^2 \times 7$$

$\Rightarrow$  The probability that two of these boxes contain exactly 2 and 3 balls is

$$= \frac{2^5 \times 3^3 \times 5 \times 7 (12+5)}{4^{10}} = \frac{3^3 \times 5 \times 7 \times 17}{2^{15}}$$

### Solution 65:

Given fog is identity function, thus

$$f(g(x)) = x$$

$$\Rightarrow f'(g(x)) \cdot g'(x) = 1$$

Put  $x = a$

$$\Rightarrow f'(g(a)) g'(a) = 1$$

$$\Rightarrow f'(b) \times 5 = 1$$

$$\Rightarrow f'(b) = \frac{1}{5}$$

**Solution 66:**

$$\text{Given, } x = \sum_{n=0}^{\infty} (-1)^n \tan^{2n} \theta \text{ and } y = \sum_{n=0}^{\infty} \cos^{2n} \theta$$

$$\Rightarrow y = 1 + \cos^2 \theta + \cos^4 \theta + \dots$$

$$\Rightarrow y = \frac{1}{1 - \cos^2 \theta} \Rightarrow \frac{1}{y} = \sin^2 \theta$$

$$\Rightarrow x = \frac{1}{1 - (-\tan^2 \theta)} = \frac{1}{y} = \sin^2 \theta$$

$$\Rightarrow \cos^2 \theta = x$$

$$\Rightarrow \frac{1}{y} + x = 1$$

$$\Rightarrow y(1 - x) = 1$$

**Solution 67:**

$$\text{Given, } \left( \frac{x}{\cos \theta} + \frac{1}{x \sin \theta} \right)^{16}$$

$$T_{r+1} = {}^{16}C_r \left( \frac{x}{\cos \theta} \right)^{16-r} \left( \frac{1}{x \sin \theta} \right)^r$$

for  $r = 8$  term is free from 'x'

$$T_9 = {}^{16}C_8 \frac{1}{\sin^8 \theta \cos^8 \theta}$$

$$T_9 = {}^{16}C_8 \frac{2^8}{(\sin 2\theta)^8}$$

$$\text{in } \theta \in \left[ \frac{\pi}{8}, \frac{\pi}{4} \right], L_1 = {}^{16}C_8 2^2 \quad \because \left\{ \text{Min value of } L_1 \text{ at } \theta = \frac{\pi}{4} \right\}$$

$$\text{in } \theta \in \left[ \frac{\pi}{16}, \frac{\pi}{8} \right], L_2 = {}^{16}C_8 \frac{2^8}{\left( \frac{1}{\sqrt{2}} \right)^8} = {}^{16}C_8 \cdot 2^8 \cdot 2^4 \quad \because \left\{ \text{min value of } L_2 \text{ at } \theta = \frac{\pi}{8} \right\}$$

$$\therefore \frac{L_2}{L_1} = \frac{{}^{16}C_8 \cdot 2^8 \cdot 2^4}{{}^{16}C_8 \cdot 2^2} = 16$$

**Solution 68:**

$$\text{Let, } I = \int \frac{d\theta}{\cos^2 \theta (\tan 2\theta + \sec 2\theta)}$$

$$\Rightarrow I = \int \frac{\sec^2 \theta}{\frac{1 + \tan^2 \theta}{1 - \tan^2 \theta} + \frac{2 \tan \theta}{1 - \tan^2 \theta}} d\theta$$

$$\Rightarrow I = \int \frac{\sec^2 \theta (1 - \tan^2 \theta)}{(1 + \tan \theta)} d\theta$$

$$\Rightarrow I = \int \frac{\sec^2 \theta (1 - \tan \theta)}{1 + \tan \theta} d\theta$$

$$\text{Put } \tan \theta = t$$

$$\Rightarrow \sec^2 \theta d\theta = dt$$

$$\Rightarrow \int \left( \frac{1-t}{1+t} \right) dt = \int \left( -1 + \frac{2}{1+t} \right) dt$$

$$= -t + 2 \log(1+t) + C$$

$$= -\tan \theta + 2 \log(1 + \tan \theta) + C$$

$$\Rightarrow -\tan \theta + 2 \log(1 + \tan \theta) + C$$

$$\Rightarrow \lambda = -1 \text{ and } f(\theta) = 1 + \tan \theta$$

**Solution 69:**

$$\text{Here, } A = \{x : x \in (-2, 2)\}$$

$$\Rightarrow B = \{x : x \in (-\infty, -1] \cup [5, \infty)\}$$

$$\Rightarrow A \cap B = \{x : x \in (-2, -1]\}$$

$$\Rightarrow A \cup B = \{x : x \in (-\infty, 2) \cup [5, \infty)\}$$

$$\Rightarrow A - B = \{x : x \in (-1, 2)\}$$

$$\Rightarrow B - A = \{x : x \in (-\infty, -2] \cup [5, \infty)\}$$

**Solution 70:**

Given system of linear equations

$$7x + 6y - 2z = 0 \quad \dots(1)$$

$$3x + 4y + 2z = 0 \quad \dots(2)$$

$$x - 2y - 6z = 0 \quad \dots(3)$$

$$\therefore \begin{vmatrix} 7 & 6 & -2 \\ 3 & 4 & 2 \\ 1 & -2 & -6 \end{vmatrix}$$

$$= 7(-20) - 6(-20) - 2(-10)$$

$$= -140 + 120 + 20 = 0$$

so infinite non-trivial solution exist

now equation (1) + 3 equation (3)

$$10x - 20z = 0$$

$$x = 2z$$

### Solution 71:

Given two A.P.'s 3, 7, 11, ....., 407 and 2, 9, 16, ....., 709

First common term = 23

common difference =  $7 \times 4 = 28$

Last term  $\leq 407$

$$\Rightarrow 23 + (n - 1) \times 28 \leq 407$$

$$\Rightarrow (n - 1) \times 28 \leq 384$$

$$\Rightarrow n \leq 13.71 + 1$$

$$n \leq 14.71$$

$$\text{So } n = 14$$

### Solution 72:

Lines must be intersecting

$$\Rightarrow (2s - 1, 4s + 3, 3s - 1) = (2t - 3, 6t - 2, \lambda t + 1)$$

$$2s - 1 = 2t - 3, 4s + 3 = 6t - 2, 3s - 1 = \lambda t + 1 \Rightarrow t = \frac{1}{2}, s = -\frac{1}{2}, \lambda = -7$$

distance of plane contains given lines from given plane is same as distance between point  $(-3, -2, 1)$  from given plane.

$$\text{Required distance equal to } \frac{|-69 + 20 - 2 + 48|}{\sqrt{529 + 100 + 4}} = \frac{3}{\sqrt{633}} = \frac{k}{\sqrt{633}} \Rightarrow k = 3$$

### Solution 73:

$$\text{Given, } |\vec{a}| = \sqrt{3}, |\vec{b}| = 5, \vec{b} \cdot \vec{c} = 10$$

$$\Rightarrow |\vec{b}| |\vec{c}| \cos\left(\frac{\pi}{3}\right) = 10$$

$$\Rightarrow 5 \cdot |\vec{c}| \cdot \frac{1}{2} = 10$$

$$\Rightarrow |\vec{c}| = 4$$

$$\text{Also, } \vec{a} \cdot (\vec{b} \times \vec{c}) = 0$$

$$|\vec{a} \times (\vec{b} \times \vec{c})| = |\vec{a}| |\vec{b} \times \vec{c}| \sin\left(\frac{\pi}{2}\right)$$

$$\sqrt{3} \times |\vec{b}| |\vec{c}| \sin \frac{\pi}{3} \times 1 = \sqrt{3} \times 5 \times 4 \times \frac{\sqrt{3}}{2} = 30$$

### Solution 74:

Given curves  $x^2 - 6x + y^2 + 8 = 0$  and  $x^2 - 8y + y^2 + 16 - k = 0$

Two circles touches each other if  $C_1 C_2 = |r_1 \pm r_2|$

Distance between  $C_2(3, 0)$  and  $C_1(0, 4)$  is either  $\sqrt{k} + 1$  or  $|\sqrt{k} - 1|$  ( $C_1C_2 = 5$ )

$$\Rightarrow \sqrt{k} + 1 = 5 \text{ or } |\sqrt{k} - 1| = 5$$

$$\Rightarrow k = 16 \text{ or } k = 36$$

$\Rightarrow$  maximum value of  $k$  is 36

**Solution 75:**

Given,  $C_0 + 5.C_1 + 9.C_2 + \dots + (101).C_{25}$

$$\Rightarrow \sum_{r=0}^{25} (4r+1) {}^{25}C_r = 4 \sum_{r=0}^{25} r \cdot {}^{25}C_r + \sum_{r=0}^{25} {}^{25}C_r$$

$$= 4 \sum_{r=1}^{25} r \times \frac{25}{r} {}^{24}C_{r-1} + 2^{25} = 100 \sum_{r=1}^{25} {}^{24}C_{r-1} + 2^{25}$$

$$= 100.2^{24} + 2^{25} = 2^{25} (50+1) = 51.2^{25}$$

So  $k = 51$